# EAT: How Far Do Raw pLM Embeddings Carry Annotations?

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# Most proteins will never be measured

**150,000,000**

protein sequences in UniProtKB [1]

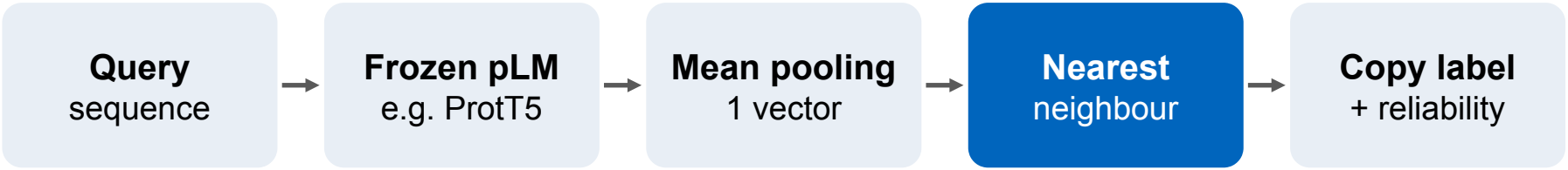
**48,401**

with experimentally measured enzyme activity [1]

→ **Everything else is annotated by transfer from a similar protein.**
Sequence similarity (BLAST, MMseqs2) fails below ~30 % identity.

[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025); counts: UniProtKB 2026_03, queried 28.10.2026.

# EAT: a query inherits its nearest neighbour's label



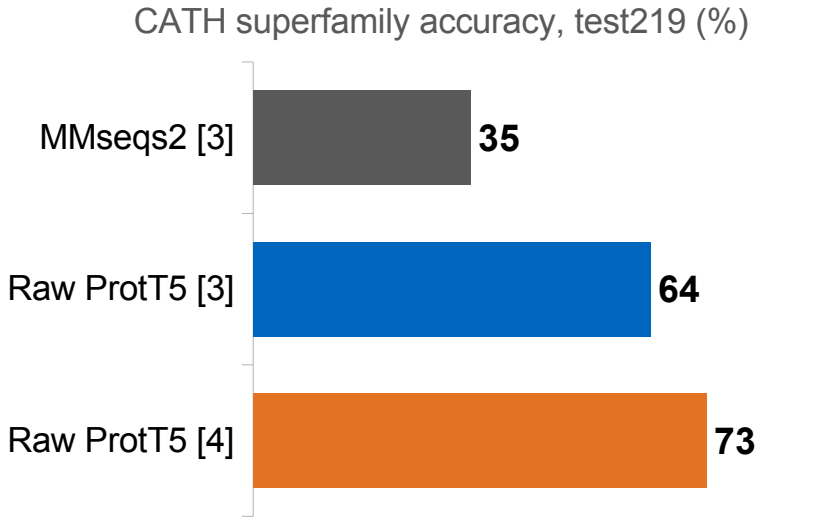

Reliability index (goPredSim): **RI = 0.5 / (0.5 + d)**

Training-free, yet beats BLAST for GO terms [2]

CATH superfamily, remote homologues: raw ProtT5 64 % vs MMseqs2 35 % [3]

[2] M. Littmann … B. Rost. Embeddings transfer GO annotations beyond homology. Sci Rep (2021).
[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

# Raw EAT: a common baseline, set up differently each time




Same pLM, same test + lookup sets: 9 points apart. No CIs reported.

- Every paper picks pLM, pooling, distance, k and lookup set differently
- The pLM matters: ProtT5 → ProstT5 raises CATH-H from 69.6 to 78.5 % [5]
- RI cutoffs were tuned for SeqVec (2021); distance scales differ across pLMs
- **No "HFSP for embeddings" [6]:** no calibrated safe-transfer threshold

[3] Heinzinger … Rost (2022). [4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model. arXiv (2023).
[5] D. Miller … C. Orengo. ContrasTED. bioRxiv (2026). [6] Y. Mahlich … Y. Bromberg. HFSP. Bioinformatics (2018).

# The labels we copy: shape (CATH) and job (EC)

Protein › Language model › Vector › Nearest neighbour › **Copy label**

## CATH: the 3D shape of a domain

domain = compact, folded building block of a protein

- **3** Class: mixed helices and sheets
- **3.40** Architecture: three-layer sandwich
- **3.40.50** Topology: Rossmann fold
- **3.40.50.300** Superfamily: shared ancestor

## EC: the reaction an enzyme performs

enzyme = protein that speeds up a chemical reaction

- **3** Class: hydrolase: cuts bonds with water
- **3.4** Subclass: cuts proteins
- **3.4.21** Sub-subclass: serine protease
- **3.4.21.4** Enzyme: trypsin

Four levels each, from general to specific: at which level does copying a label still work?

[7] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021). [8] IUBMB. Enzyme Nomenclature (ExplorEnz).
Data: [3], [6], UniProt [1].

# A language model turns each protein into a point in space

Protein › **Language model** › **Vector** › Nearest neighbour › Copy label

**M K T A Y I A K Q R Q I S F V K …**
protein = chain of amino acids

↓

**Protein language model**
has read millions of sequences

↓

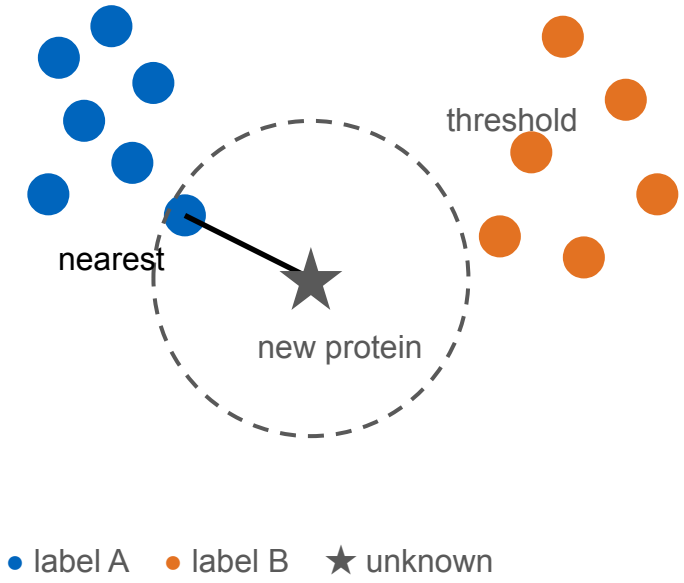
**[ 0.12, −0.80, 0.33, … ]**
≈ 1,000 numbers = one point

- Learns the “grammar” of proteins, like a text model learns a language
- Similar proteins end up close together
- We compare 11 models from 4 families (ProtT5, ESM-2, ESM C, Ankh)
- **Each model builds its own space:** so each needs its own threshold
- **Nothing is trained:** vectors are computed once per model

[9] A. Elnaggar … B. Rost. ProtTrans. IEEE TPAMI (2022). [10] Z. Lin … A. Rives. ESM-2. Science (2023). [4] Elnaggar … Rost. Ankh (2023).

# Copy the label of the closest protein – if it is close enough

Protein › Language model › Vector › **Nearest neighbour** › Copy label




- **Nearest neighbour:** the known protein at the smallest distance
- **Distance:** straight line (Euclidean) or angle between vectors (cosine)
- **Threshold:** too far away → do not copy the label
- **Goal:** one threshold per model and level, e.g. “below this distance 95 % of copied labels are correct”

[2] Littmann … Rost (2021).

# How we test it: reproduce, compare, scale up

Protein › Language model › Vector › Nearest neighbour › Copy label

## 1 Reproduce

Rerun a published result: ProtT5 on CATH

84 / 67 / 57 / 64 % per level

*same numbers → our code is right*

## 2 Compare

Copied label vs true label, per level

vs the classic way: copy from the most similar sequence

*where do embeddings win?*

## 3 Scale up

11 models × 2 distances

CATH and EC

*which model, which distance, which threshold?*

**Fair test: tune on one part of the data, report on another**

| **Known proteins** | **Tuning set** | **Test set** |
|---|---|---|
| we copy labels from | choose distance, set thresholds | used once, for final numbers |

[3] Heinzinger … Rost (2022): published ProtT5 result on CATH test219.

# What we will deliver

**1 Best model and distance**
for shape (CATH) and job (EC), and where it beats sequence similarity

**2 A safe distance**
per model and level, e.g. ≥ 95 % of copied labels correct

**3 One trust score**
same meaning for every model: 0.9 = 90 % of such labels correct

**4 Open benchmark**
code + fixed data splits: test any new model the same way

→ Thresholds go into **ProtSpace**, the lab's embedding viewer. First step (23.11): reproduce ProtT5.

## Thank you – questions?

# References


[1] UniProt Consortium. UniProt: the Universal Protein Knowledgebase. Nucleic Acids Res (2025).

[2] M. Littmann … B. Rost. Embeddings from deep learning transfer GO annotations beyond homology. Sci Rep (2021).

[3] M. Heinzinger … B. Rost. Contrastive learning on protein embeddings enlightens midnight zone. NAR Genom Bioinform (2022).

[4] A. Elnaggar … B. Rost. Ankh: Optimized protein language model unlocks general-purpose modelling. arXiv (2023).

[5] D. Miller … C. Orengo. ContrasTED: Contrastive domain embeddings for remote homology classification. bioRxiv (2026).

[6] Y. Mahlich … Y. Bromberg. HFSP: High speed homology-driven function annotation of proteins. Bioinformatics (2018).

[7] I. Sillitoe … C. Orengo. CATH: increased structural coverage of functional space. Nucleic Acids Res (2021).

[8] IUBMB. Enzyme Nomenclature. ExplorEnz database, www.enzyme-database.org.

[9] A. Elnaggar … B. Rost. ProtTrans: Toward understanding the language of life. IEEE TPAMI (2022).

[10] Z. Lin … A. Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. Science (2023).

Project description: Chair of Bioinformatics (I12), TUM. EAT: A benchmark for annotation transfer with raw pLM embeddings (2026).